Helium-Neon Laser Advanced Lab Course

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Contents

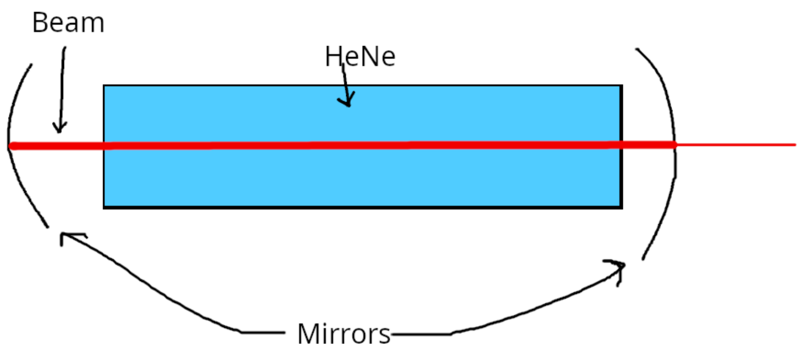

Figure 1: Basic laser resonator

1 Abstract

The goal of this experiment is to investigate the properties of a Helium-Neon Laser. To do this, the Gaussian beam of the laser will be measured using the scanning knife edge and clip level methods. The resonator's stability will be evaluated by adjusting the mirrors. The goal is to use a Michelson interferometer to detect coherence length and suppress the basic Gaussian mode, resulting in greater transverse modes than usual red and orange laser photons.

2 Preparation

2.1 Gaussian fundamental mode

A laser resonator looks like in figure 1. A beam which is generated in it can be described as

$$E(x, y, z) = u(x, y, z)exp(-ikz) \tag{1}$$

when it is aligned to the z direction and the intensity is decreasing in z direction. It must satisfy the paraxial wave equation:

$$\partial_x^2 u + \partial_x^2 u - 2ik\partial_z u = 0 \tag{2}$$

The most simple solution to this is the *Gaussian fundamental mode*

$$E_{00}(x, y, z) = \sqrt{\frac{2}{\pi}} \frac{q(0)}{\omega(0)q(z)} exp\left(-ik\frac{x^2 + y^2}{2R(z)} + i\Psi(z) - ikz\right) \tag{3}$$

which has the following properties:

1. **Beam radius:**
$$\omega(z) := \omega_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2} \tag{4}$$
The beam radius is the distance from the z axis, where the intensity has fallen to $1/e^2$ of the initial value. The *beam waist* ω_0 is the radius at the origin $z = 0$.

2. **Rayleigh length:**
$$z_R := \frac{\pi\omega_0^2}{\lambda} \tag{5}$$
The area $z \in [-z_R, +z_R]$ is called *confocal area*. In there, the beam parameters change the most.

3. **Radius of curvature:**
$$R(z) := z + \frac{z_R^2}{z} \tag{6}$$
It is the radius of curvature of the spherical waves.

4. **Gouy-Phase**
$$\frac{i}{q(z)} := \frac{\lambda}{\pi\omega^2(z)} + \frac{i}{R(z)} := \frac{exp(i\Psi(z))}{|q(z)|} \tag{7}$$
In the confocal area, a phase shift of $\frac{\pi}{2}$ passes through and between $-\infty$ and ∞ a phase shift of π. Higher modes have a higher Gouy phase shift.

5. **Divergence:**
$$\Theta_{Div} := \arctan\left(\frac{\omega(z)}{z}\right) = \arctan\left(\frac{\omega_0}{z_R}\right), \quad \text{if} \quad z \gg z_R \tag{8}$$
The divergence is the opening angle of the beam. It comes when z is far away from the confocal area.

These properties are shown in figure 2.

The resonator of this experiment consists of a plane mirror at $z = 0$ and a spherical mirror with curvature radius of $\rho = 700$mm. The beam exists, if the radius of curvature at the position of the spherical mirror matches the mirror curvature radius (equation 3):
$R(z) = z + \frac{z_R^2}{z} \stackrel{!}{=} \rho \rightarrow z_R = \sqrt{(\rho - z)z} = \sqrt{(700\text{mm} - z)z}$

The resonator length is $z = 50$cm with a light wavelength of $\lambda = 632.8$nm. As derived from Equation 2, the relationship is expressed as follows:

$$z_R = \frac{\pi\omega_0^2}{\lambda} \rightarrow \omega_0 = \sqrt{\frac{z_R\lambda}{\pi}} = \sqrt{\frac{\lambda\sqrt{(\rho - z)z}}{\pi}} = \sqrt{\frac{\lambda}{\pi}} \cdot \sqrt[4]{(\rho - z)z} = 252.38\mu\text{m} \tag{9}$$

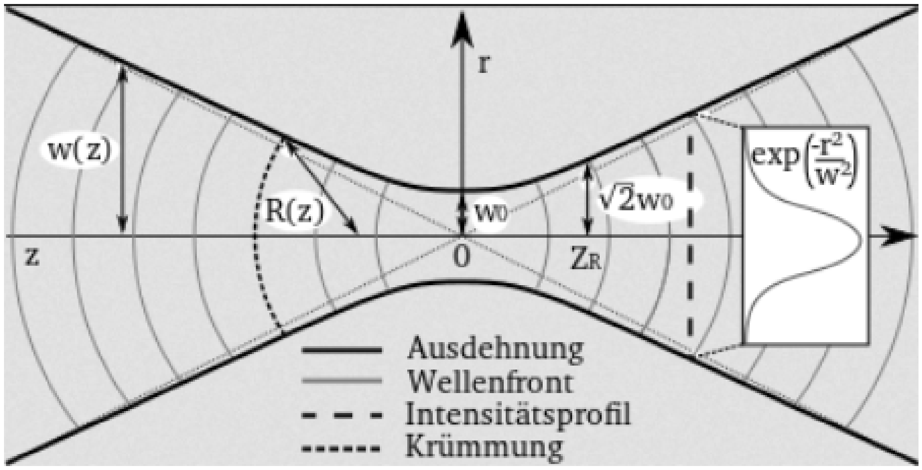

Figure 2: Gaussian fundamental mode

This results in a Rayleigh length of:
$z_R = \sqrt{(700\text{mm} - 50\text{cm}) \cdot 50\text{cm}} = 316.23\text{mm}$

The radius of curvature becomes minimal at the distance, where the derivative of it is zero:
$\frac{dR(z)}{dz} = 1 - 2 \cdot \frac{z_R^2}{z^2} \stackrel{!}{=} 0 \rightarrow z = \sqrt{2} \cdot z_R$

The maxima are at $z = 0$ and $z \rightarrow \infty$, because at the plane mirror there is a plane wave and very far away the radius of the spherical waves grows to infinity. The laboratory is 3m long and a detector is placed at the wall. The radius of the beam there is (equation 1):
$\omega(3\text{m}) = \omega_0 \sqrt{1 + \left(\frac{3\text{m}}{z_R}\right)^2} = 2.41\text{mm}$

The available detector has a diameter of $d = 3.5\text{mm}$, it is smaller than the beam diameter $2\omega(3\text{m}) = 4.72\text{mm}$, so the detector is too small to measure most of the beam. To focus the beam, a converging lens can be placed in the beam.

2.2 Stability of the resonator and g-parameters

The stability of a laser resonator is described with the g-parameters, which are $g_1 = 1 - \frac{L}{R_1}$ and $g_2 = 1 - \frac{L}{R_2}$. The must fulfill the condition:

$$0 \leq g_1 \cdot g_2 \leq 1 \tag{10}$$

The g-parameters are derived from matrix optics. A beam in ray optics can be described as $\vec{a}_i := (r_i, \theta_i)$ with r_i as the distance to the optical axis and θ_i as

the angle.
An optical element in paraxial ray optics can be described as an ABCD-matrix M, so the beam is linear transformed: $\vec{a}_2 = M \cdot \vec{a}_1$
The matrix for a spherical mirror is $M_{R_i} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_i} & 1 \end{pmatrix}$
and the matrix of the propagation through a space with length L is
$M_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$.

The matrix of a laser resonator is then $M = M_{R_1} \cdot M_L \cdot M_{R_2} \cdot M_L$
The eigenvalues of the matrix are $\lambda_{1,2} = \frac{\mathrm{Tr}(M)}{2} \pm \sqrt{\left(\frac{\mathrm{Tr}(M)}{2}\right)^2 - 1}$.

The discriminant generates two different cases for the trace of M:
1. $|\mathrm{Tr}(M)| > 2$: The eigenvalues are real and the magnitude of the beam vectors increase every circulation in the resonator. The resonator is *unstable.*
2. $|\mathrm{Tr}(M)| \leq 2$: The eigenvalues are complex conjugated to each other, and they have $|\lambda_i| = 1$. The resonator is *stable.*

$$Tr(M) = A + D = 2 - 8L\left(\frac{1}{R_1} + \frac{1}{R_2}\right) + \frac{4L^2}{R_1 R_2}$$

If we define the g-parameters:

$$g_1 = 1 - \frac{L}{R_1}$$

$$g_2 = 1 - \frac{L}{R_2}$$

we can rewrite the **stability-condition** as: $0 \leq g_1 \cdot g_2 \leq 1$
and reform back:

$$4g_1g_2 \leq 4$$

$$4g_1g_2 - 2 \leq 2$$

$$Tr(M) \leq 2$$

Taking the border-condition for stability:
To obtain a stable resonator, the equation $|\lambda_1| = |\lambda_2| = 1$ has to be fulfilled. Using the g-parameters from the beginning of the subsection. The diagonal values of the matrix M are $A = 1 - \frac{2L}{R_2}$ and $D = -\frac{4L}{R_1} + \frac{4L^2}{R_1 R_2}$ and it follows that $\mathrm{Tr}(M) = A + D = 4g_1 \cdot g_2 - 2$. This condition is equivalent to equation 10, because

$$\begin{aligned} 0 \leq g_1 \cdot g_2 \leq 1 &\Leftrightarrow g_1 g_2 \in [0,1]_{\subseteq \mathbb{R}} \\ \Leftrightarrow 4g_1g_2 \in [0,4] &\Leftrightarrow 4g_1g_2 - 2 \in [-2,2] \\ \Leftrightarrow \mathrm{Tr}(M) \in [-2,2] &\Leftrightarrow |\mathrm{Tr}(M)| \leq 2 \\ &\Leftrightarrow |\lambda_1| = |\lambda_2| = 1 \end{aligned} \tag{11}$$

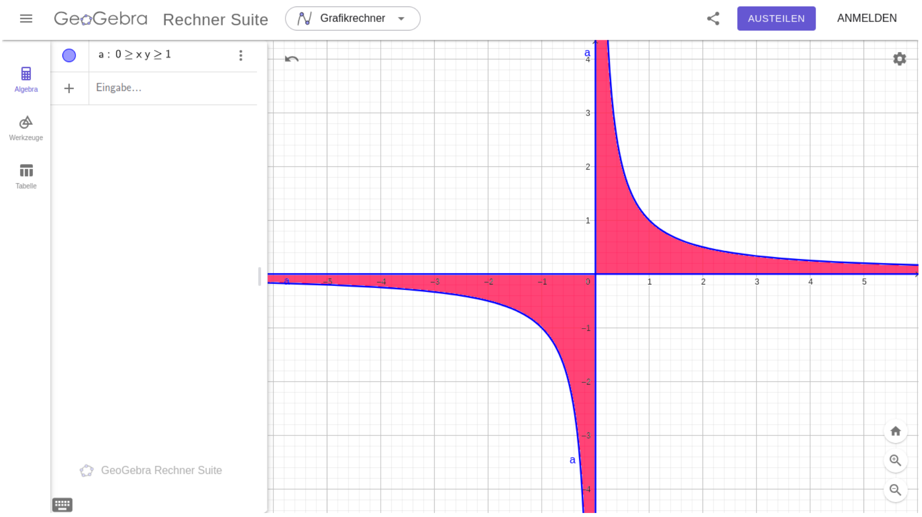

Figure 3: Stability of a resonator: the pink area hold a stable resonator, where $x = g_1$ and $y = g_2$

In figure 3 the stability condition with the g-parameters is shown. There are several types of resonators. Three are discussed here. The planar, concentric and confocal.

A planar resonator is has two planar mirrors with curvature ∞, so its g-parameters are $g_1 = g_2 = 1$.
A concentric resonator has two mirrors with curvature radius $R_{1,2} = \frac{L}{2}$ and g-parameters $g_1 = g_2 = -1$.
A confocal resonator has two mirrors with curvature radius $R_{1,2} = L$ and g-parameters $g_1 = g_2 = 0$.

The planar resonator is not very stable, because it is at the g-parameters $(0, 0)$ and every small error goes outside the stable region. These three differen resonator types are shown in figure 4. The concentric and confocal are more stable and the tolerance is higher, because there is more stable area in the neighbourhood of the points $(-1, -1)$ and $(1, 1)$. The most stable resonators have g-parameters inside the stable area, for example $(0.5, 0.5)$, so the curvature radii are bigger than the resonator length. Then there is a stable Gaussian beam shape.

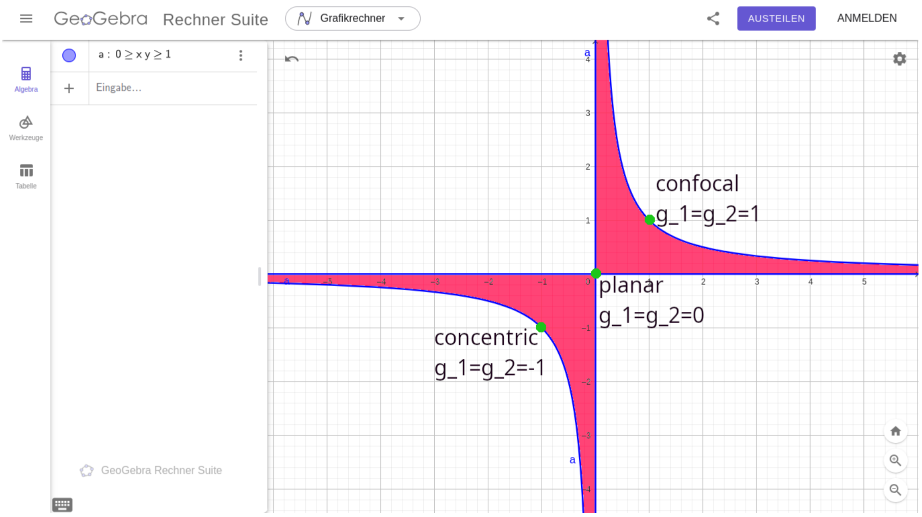

Figure 4: The green dots describing different kinds of resonators

To estimate the stability condition of the resonator used in the experiment, the following parameters are used:

Given Parameters:

- Planar mirror: $R_1 = \infty$ (because a planar mirror can be considered as having an infinite radius of curvature).
- Spherical mirror: $R_2 = 700\,\text{mm}$.
- Resonator length: L.

Calculating the g-Parameters: 1. **For the planar mirror $R_1 = \infty$:**

$$g_1 = 1 - \frac{L}{R_1} = 1 - \frac{L}{\infty} = 1$$

2. **For the spherical mirror $R_2 = 700\,\text{mm}$:**

$$g_2 = 1 - \frac{L}{700\,\text{mm}}$$

Stability Condition: For the resonator to be stable:

$$0 < g_1 \cdot g_2 < 1$$

Substituting $g_1 = 1$ and $g_2 = 1 - \frac{L}{700}$:

$$0 < 1 \cdot \left(1 - \frac{L}{700}\right) < 1$$

This simplifies to:

$$0 < 1 - \frac{L}{700} < 1$$

Breaking Down the Stability Condition: 1. **Lower Bound:**

$$0 < 1 - \frac{L}{700}$$

This implies:

$$\frac{L}{700} < 1 \quad \text{or} \quad L < 700\,\text{mm}$$

2. **Upper Bound:**

$$1 - \frac{L}{700} < 1$$

This implies:

$$\frac{L}{700} > 0 \quad \text{or} \quad L > 0\,\text{mm}$$

Conclusion: The resonator is stable when the length L satisfies:

$$0 < L < 700\,\text{mm}$$

This means the resonator will be stable for any length L between 0 and 700 mm (excluding exactly 0 and 700 mm).

1. Calculation of Free Spectral Range (FSR)

The Free Spectral Range (FSR) is the frequency difference between two neighboring longitudinal modes in a resonator. It is given by the formula:

$$\text{FSR} = \delta\nu = \frac{c}{2L}$$

where:

- c is the speed of light in a vacuum, approximately 3×10^8 m/s,
- L is the length of the resonator.

Given:

- Wavelength $\lambda = 632.8\,\text{nm}$,
- Resonator length $L = 60\,\text{cm} = 0.60\,\text{m}$.

Substituting these values into the FSR formula:

$$\delta\nu = \frac{3 \times 10^8\,\text{m/s}}{2 \times 0.60\,\text{m}} = \frac{3 \times 10^8}{1.2}\,\text{Hz} = 2.5 \times 10^8\,\text{Hz} = 250\,\text{MHz}$$

So, the Free Spectral Range (FSR) of the Helium-Neon Laser is **250 MHz**.

2.3 2. Spectrometer Resolution to Determine Single Mode Operation

For the spectrometer to resolve whether the laser is operating in a single mode, it must be able to distinguish between adjacent modes. Therefore, the resolution $\delta\nu$ of the spectrometer must be less than or equal to the Free Spectral Range (FSR) calculated above. Thus, the spectrometer should have a resolution of at least **250 MHz** or better. This means the spectrometer must be capable of resolving frequency differences smaller than 250 MHz to detect if the laser is operating in single mode.

3 Experiment

3.1 Scanning Knife Edge

The first step is to verify if the laser mode is the Gaussian fundamental mode. The beam is projected to the wall which is 3m away from the laser, so the beam is widened enough to see it with the eye. The Gaussian fundamental mode has a 2D Gaussian profile as described in figure 2. The goal is to get the beam profile from power measurements. In every measure step, a razor blade is placed a few micrometres more inside the beam, so that a growing part of the beam is blocked. The starting point of the razor blade is outside the beam, and the end is where the entire beam is blocked. The measurement is made at three different beam distances. The result is shown in 6. The data of an exact Gaussian beam is the mirrored error function. This is shown with a simulation of the knife edge method with a perfect Gaussian profile in figure 5. The reason for the error function is, that the error function is the antiderivative of the Gaussian function and the measured voltage of the photodiode is the integration over the Gaussian beam profile, so the integration results in the error function. The real measured data deviates from the error function, so it is likely that there are higher order modes inside the beam.

In figure 6 the measured voltages from the scanning knife edge method are shown. The beam width is defined at the distance, where the power of the beam is 1/e of the maximum power. For the Gaussian distribution, the value falls down to 1/e at the distance $\sqrt{2}\sigma$ from the expected value. So the beam width is given with $\omega = \sqrt{2}|\sigma|$. The absolute value is used, because σ is negative to invert the slope of the error function fit.

The beam widths are:
$\omega(92.5\text{cm}) = 97.95\mu\text{m}$
$\omega(152.0\text{cm}) = 107.79\mu\text{m}$
$\omega(212.5\text{cm}) = 150.97\mu\text{m}$

$\omega_0 = 80\mu\text{m}$, $\omega(z_R) = \sqrt{2}\omega_0 \approx 113.14\mu\text{m}$

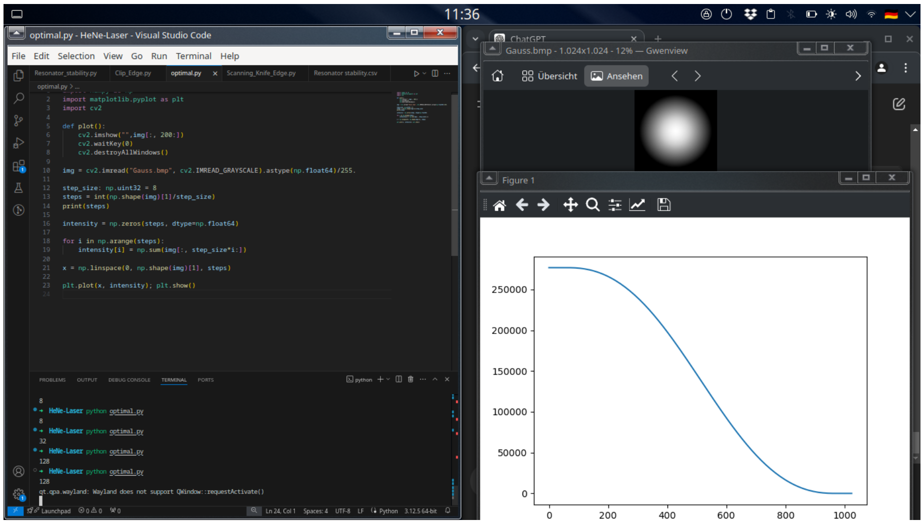

Figure 5: Simulation of the knife edge method with a perfect Gaussian profile. The left window is the code. It cuts the beam in every calculation step, like the knife in reality. The upper window is the 2D Gaussian image, the lower window is the estimated data.

To get z_R the graphical solution is estimated with figure 7.
$z_R = 146.42$cm

The first measurement at 92.5cm is in the near field and the second and third are in the far field. The divergence angle for the geometric approximation is closest to the last value which is farthest away from the resonator. The fit function for the divergence angle can be estimated at a very far point. For this estimation, the angle value at $z = 5$m is used which is $\Theta(\omega(5\text{m}), 5\text{m}) = 90.64\mu$rad

The values for z_R, and ω_0 are significantly different than in theory (see section 2.1), it is likely that there are big measurement or calculation errors. In theory the parameters are $z_R = 316.23$mm and $\omega_0 = 252.38\mu$m.

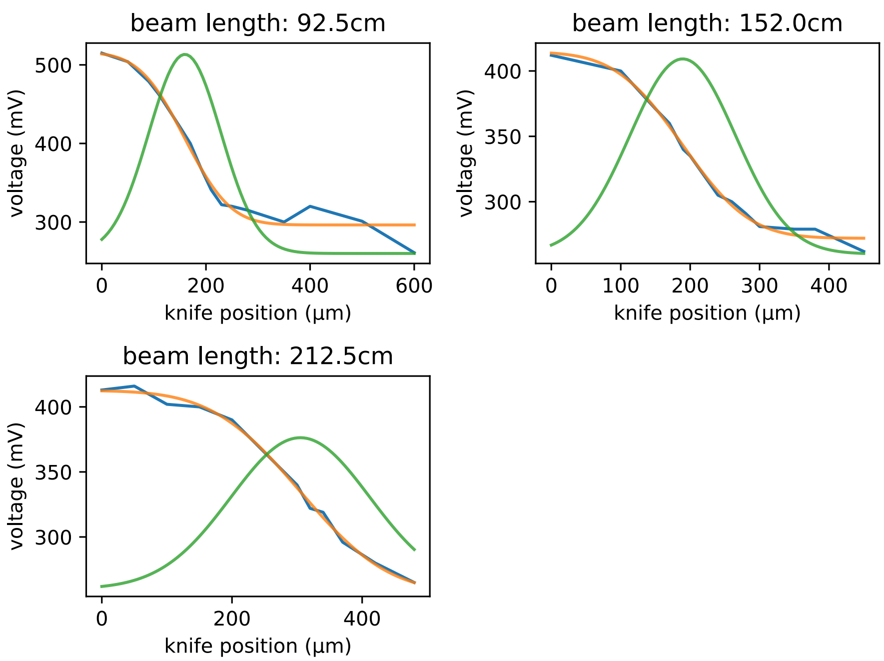

Figure 6: Scanning Knife Edge data for the different beam lengths.
blue: measured data, orange: error function fit, green: derived Gaussian beam

Fit parameters [A (mV), z0 (µm), σ (µm), U0 (mV)]:
l=92.5cm: [219.95 159.54 -69.26 296.29]
l=152.0cm: [142.57 189.26 -76.22 272.03]
l=212.5cm: [155.62 305.04 -106.75 257.04]

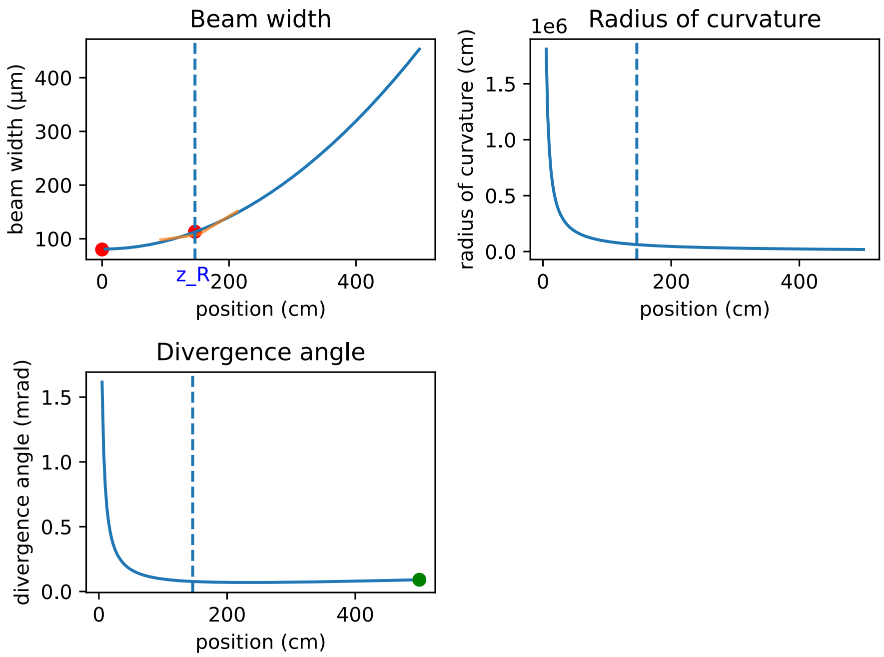

Figure 7: Beam shape with graphically estimated Rayleigh length and beam waist from the scanning knife edge method. The orange line in the beam width subplot is the measurement data and the green dot in the divergence angle subplot is the estimated value for the geometric divergence angle.

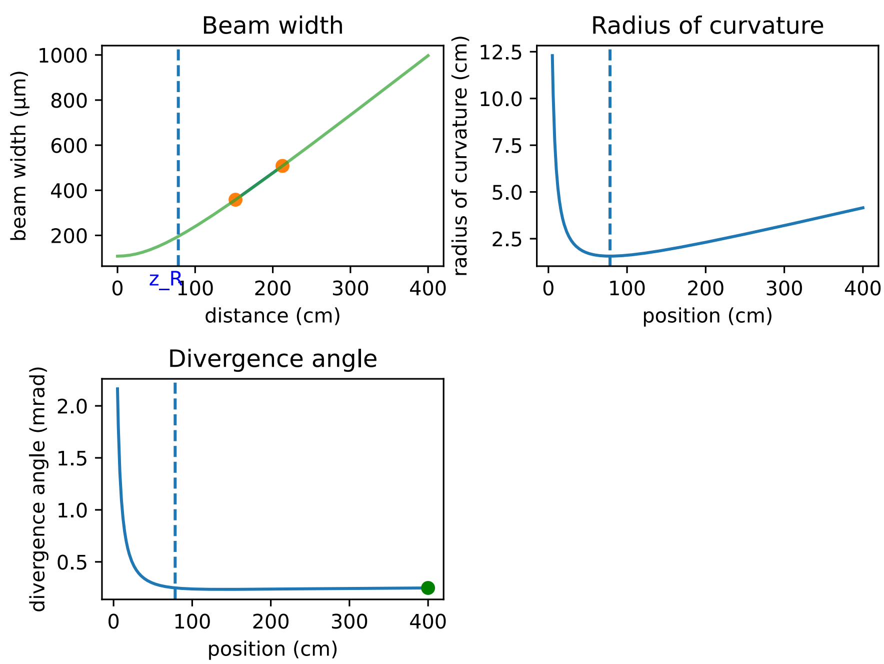

Figure 8: Beam fits extracted from the clip level method, the beam width is shifted because it would be negative else. The dotted vertical line is the Rayleigh length.

3.2 Clip Level

The clip level method takes the knife positions, where the power of the beam decreases to $1-\epsilon$ and ϵ of the original power. The distance between these knife positions is D_c and the beam width of the Gaussian fundamental mode is 2σ. The beam power profile is given by

$$\epsilon = \frac{1}{2}\left(1 - \text{erf}\left(\frac{D_c}{2\sqrt{2}\sigma}\right)\right) \tag{12}$$

To get the beam width $2\sigma = \omega(z)$ the equation is formed to

$$\omega(z) = 2\sigma = \frac{D_c}{\sqrt{2}erf^{-1}\left(1-2\epsilon\right)} \tag{13}$$

In this experiment, $\epsilon = 0.1$ is used.

Beam length (cm)	Max (mV)	Min (mV)	10% interpolation (mV)	90% interpolation (mV)	10% position (µm)	90% position (µm)
152	430	260	277	143	89	277
212.5	416	260	276	400	110	490

Table 1: Clip level measurements

In table 1 the voltages for the ϵ powers and the knife positions are shown. The distances D_c of the positions are
$|277-89|\mu\text{m} = 188\mu\text{m}$ for beam length 152.0cm and
$|490-110|\mu\text{m} = 380\mu\text{m}$ for beam length 212.5cm.

These values put in in equation 13 give
$\omega(152.0\text{cm}) = 2\sigma = \frac{188\mu\text{m}}{\sqrt{2}erf^{-1}(1-0.2)} = 146.7\mu\text{m}$

$\omega(212.5\text{cm}) = 2\sigma = \frac{380\mu\text{m}}{\sqrt{2}erf^{-1}(1-0.2)} = 296.5\mu\text{m}$

In figure 8 the beam width, radius of curvature and angle are shown. It deviates from the theory. It is likely that there are unprecise measurements in it, also the count of measurement points is too small. Four points in the near field and two in the far field are a valid count of measurement points. But only two far field points are estimated. The Rayleigh length estimated from the clip level method is $z_R = 78.33\text{cm}$ and the beam waist is $\omega_0 = 211.50\mu\text{m}$. The Rayleigh length deviates significantly from the theory value of $z_R = 316.23\text{mm}$. The theoretical beam waist $\omega_0 = 252.38\mu\text{m}$ is also deviating from the estimated value.

3.3 Stability

In this experiment, the stability area of the resonator is investigated. As described in the preparation sub section *Stability of the resonator*, the g-parameters of the planar concave resonator which is used are $g_1 = 1$ and $g_2 = 0$. In figure 9

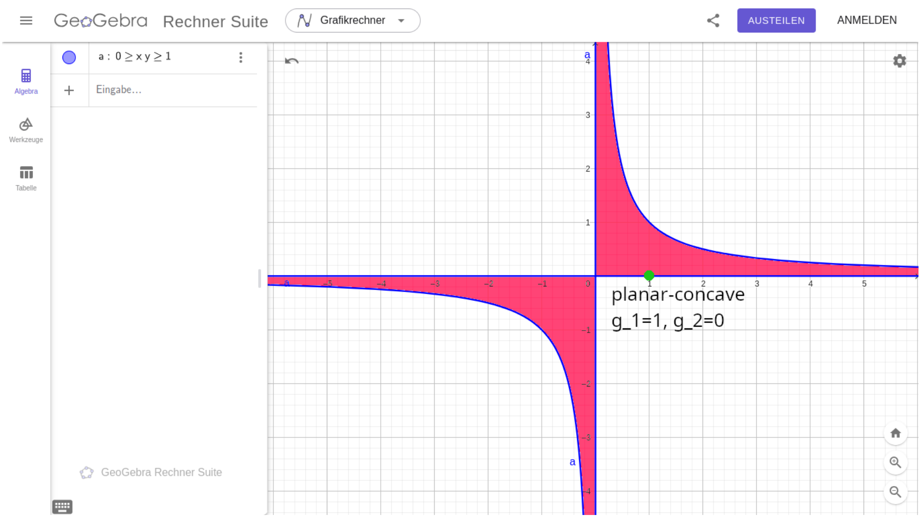

Figure 9: resonator stability diagram with g-parameter of the planar-concave resonator

the g-parameters are shown in the stability diagram. The resonator is stable at a length of 70cm and has a high output power. In figure 10 it is shown that the resonator is stable and has output power when it is larger than 70cm. The g_2-parameter would be about -0.08. The reason is, that the spherical mirror has no exact radius of curvature of 70cm. It has slightly more because the manufacture has tolerance levels. It has a curvature radius that is slightly higher than 75.5cm, because it still has output power. The radius of curvature is the length of the resonator, where the beam completely disappears and the measured voltage is the dark voltage of 267mV. This is the reason, that the g-parameter is not valid, the true g_2-parameter cannot be lower than 0 at 75.5cm resonator length.

3.4 Coherence length

To estimate the approximate coherence length, a Michelson interferometer is placed inside the beam. The interferometer used is shown in figure 11. The beam is interfered with itself but with different distances from the mirrors to the beam splitter, so the waylength of the light is different. At the beginning of the experiment, the length of both ways is 4.5cm. The interference pattern of the matched beam parts is very large. It is larger than the beam itself. The second mirror is moved and the pattern has a decreasing size (its frequency is increasing), also the pattern changes between horizontal, vertical, parabolic and irregular patterns. The mirror is moved further away slowly. At 15cm distance, the interference pattern fades away. The interference patterns only appear, when the mixing beam parts have a constant wave pattern to each other. The length

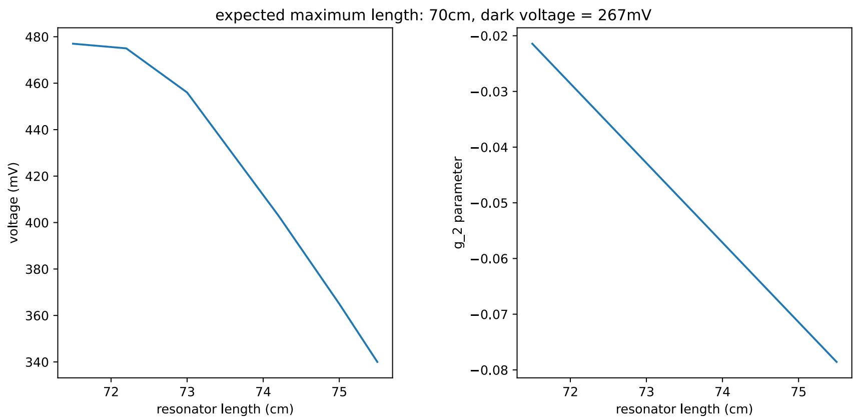

Figure 10: Output power of the resonator with over the resonator length (left) and the g-parameter of the spherical mirror over the resonator length (right)

of the constant patterns is called the *coherence length*. The coherence length is the distance difference between the mirrors doubled, because this is the path length which is propagated additionally from the second beam compared to the first beam. The double is appearing because the beam has to propagate to the mirror and also back to the beam splitter, so the way is double the distance. Because the first mirror is 4.5cm and the second mirror is 15cm away from the beam splitter, the approximate coherence length is $2 \cdot |4.5\text{cm} - 15\text{cm}| = 21\text{cm}$.

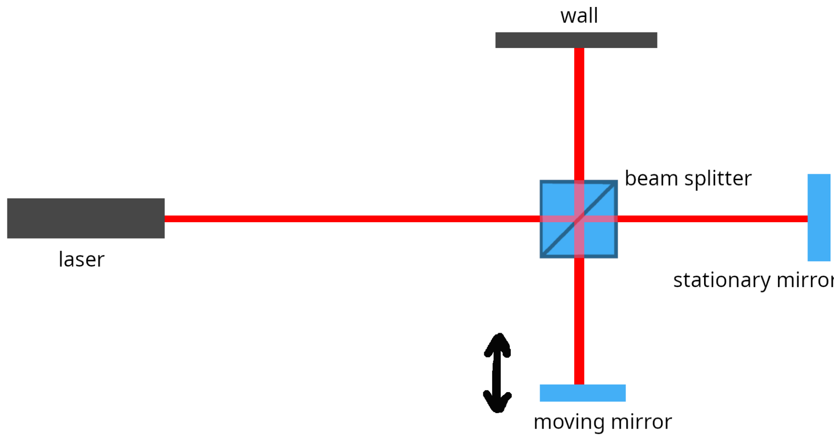

Figure 11: Michelson interferometer, the beam splitter and the mirrors generate an interference of the laser with itself but with a different path length

References
1) Experiment instructions
http://www.fp.fkp.uni-erlangen.de/fortgeschrittenenpraktikum/ versuchsangebot-fuer-bsclanf/bsc-versuchsgruppe-d.shtml (status: 15/10/2019)

(The other figures and theoretical knowledge are self made.)

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{dt}}\Psi(\mathrm{x},\mathrm{y},\mathrm{t}) = -\frac{\hbar^2}{2m}\left(\frac{\mathrm{d}^2}{\mathrm{dx}^2}\Psi(\mathrm{x},\mathrm{y},\mathrm{t}) + \frac{\mathrm{d}^2}{\mathrm{dy}^2}\Psi(\mathrm{x},\mathrm{y},\mathrm{t})\right) + V(\mathrm{x},\mathrm{y})\Psi(\mathrm{x},\mathrm{y},\mathrm{t})$$